

MKD Slidea — Technical Documentation & Investment Proposal

1. Executive Summary
2. Problem & Market
3. Product Overview
4. Technical Architecture
5. Traction & Metrics
6. Business Model
7. Financial Plan (3-Year Projection)
8. The Ask — Alibaba Cloud (Combined Proposal)
9. Roadmap (next 12 months)
10. Risks & Mitigation
11. Team
12. Appendix

MKD Slidea — Technical Documentation & Investment Proposal

Prepared for: Alibaba Cloud — Investment, Cloud Infrastructure & Strategic Partnership
Company: MKD Slidea (slidea.mismatch.net) — Skopje, North Macedonia **Founder:** Igor Bogdanoski (full-stack engineer & product owner) **Date:** July 2026 **Document version:** 1.0

1. Executive Summary

MKD Slidea is a production-ready, Macedonian-language interactive presentation and audience-response platform — a localized alternative to Mentimeter, Kahoot! and Slido, purpose-built for the North Macedonia education system and aligned with the national **BRO curriculum**. It lets teachers, trainers and event hosts run live polls, quizzes, word clouds, Q&A and surveys where audiences respond in real time from any phone, with results projected live.

The product is **live in production**, technically mature, and operating at a fraction of the cost of Western competitors because it runs on a **self-hosted open-source stack** (Supabase on a single VPS) with **Google Gemini AI** powering question generation, auto-grading and curriculum-aware semantic search.

The opportunity: the Western Balkans (North Macedonia, Albania, Serbia, Kosovo, Bulgaria, Croatia — ~20M people, ~5M students) has no dominant localized audience-response platform. Global tools (Mentimeter, Kahoot!) are English-first, expensive, and not aligned with local curricula. MKD Slidea has the technical foundation, the curriculum data, and the multilingual architecture to capture this market.

The ask (combined, investor-selectable): 1. **Cloud infrastructure support** — Alibaba Cloud credits / sponsored infrastructure to scale from a single VPS to a regional, highly-available deployment. 2. **Direct seed investment** — €25,000–€100,000 to fund regional expansion, i18n completion and enterprise features. 3. **Strategic partnership** — integration with Alibaba Cloud AI (Qwen / Model Studio), listing on the Alibaba Cloud Marketplace, and joint go-to-market for the CEE/Balkans education and enterprise segments.

2. Problem & Market

2.1 The problem

- Teachers and trainers in North Macedonia and the wider Balkans lack an affordable, **native-language** tool for interactive, real-time classroom engagement.
- Global alternatives (Mentimeter ~€100+/yr, Kahoot! Pro, Slido) are priced in EUR/USD for Western markets, English-first, and **not aligned with the local BRO curriculum**.

- Institutional procurement (schools, universities, NGOs, corporate training) needs local invoicing, GDPR compliance, and curriculum relevance — none of which global tools provide out of the box.

2.2 Target market

Segment	Description	Willingness to pay
K-12 teachers	~25,000 in North Macedonia; ~500,000 across the Balkans	Low (freemium → Pro)
Universities & faculties	UKIM, DUI, FINKI, regional universities	Medium (departmental licenses)
Corporate trainers / NGOs	Team workshops, trainings, conferences	Medium-High (event & seat licenses)
Schools & institutions	Centralized procurement, branding, admin control	High (institutional licenses)

2.3 Competitive positioning

Feature	MKD Slidea	Mentimeter	Kahoot!
Native Macedonian + Balkan languages	✓ (7 locales architected)	✗	✗
BRO curriculum alignment + AI generation	✓	✗	✗
Self-hosted / data sovereignty option	✓	✗	✗
8 activity types (incl. ranking, scale, survey)	✓	Partial	Partial
Free tier (200 participants)	✓	Limited	Limited
Pro price	€20/year	~€100+/year	~€60+/year

3. Product Overview

3.1 Core capabilities (all live in production)

- **8 interactive activity types:** multiple-choice poll, quiz (with leaderboard & instant feedback), word cloud, open text, star rating, ranking (Borda-count scoring), 1–10 scale, and multi-question

survey forms.

- **Live Presenter view:** real-time animated results (bars / donut / podium / numbers), projected to the audience; live participant counter via presence tracking; flying emoji reactions; Q&A panel with upvoting, moderation and pinning.
- **Phone-as-remote-control:** the host drives the presentation from their phone (next/previous, lock voting, reset) while it projects on screen — verified end-to-end.
- **AI-powered authoring (Google Gemini):** curriculum-aligned question generation using Bloom’s taxonomy, image/PPTX import (OCR + analysis), AI auto-grading of open answers, AI insights per session, and AI session-recap emails.
- **RAG semantic search:** 228 Macedonian curriculum chunks embedded (gemini-embedding-2, 768-dim) for curriculum-aware suggestions.
- **Templates:** 39 curated starter templates across 13 subjects (Math, Physics, Chemistry, Biology, Geography, History, Macedonian, English, CS, Art, Music, PE, Business) + a public **community template gallery** with SEO-optimized detail pages.
- **Multilingual:** Macedonian (full) + architecture for Albanian, Serbian, Croatian, Bulgarian, Romanian, English (hreflang + locale files).
- **SEO:** build-time prerendering, sitemap generation, JSON-LD structured data (Course, FAQPage, EducationalApplication), per-template OG images (satori), 7-language hreflang alternates.
- **Operations:** event scheduling & async mode, co-host access, custom branding (color/font/logo), public results pages, CSV/PDF/Markdown export, web push notifications (VAPID), transactional email (Resend).
- **Legal:** GDPR-compliant Privacy Policy and Terms of Service (North Macedonia law).

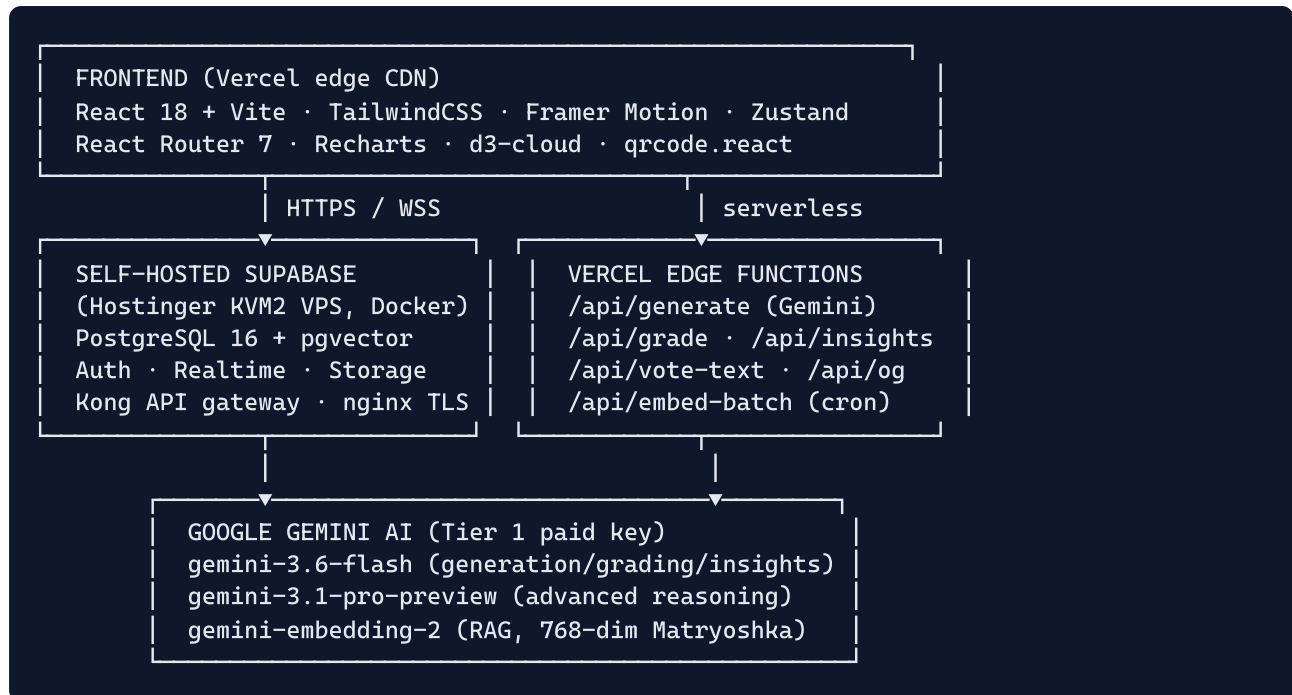
3.2 Product maturity scores (internal audit, June 2026)

Category	Score
SEO	95/100
Functionality	92/100
Security	88/100
SaaS features	78/100
Accessibility	78/100
i18n coverage	35/100 (next expansion lever)

The application is at commercial grade for the Macedonian market. The next major milestone for regional expansion is completing i18n (Albanian, Serbian).

4. Technical Architecture

4.1 Stack overview



4.2 Key technical decisions

- **Self-hosted Supabase (open source)** on a Hostinger KVM2 VPS (2 vCPU, 7.8 GB RAM, Docker) instead of paid Supabase Cloud — **data sovereignty** (all data in the EU/region) and near-zero marginal database cost. PostgreSQL 16 with the **pgvector** extension powers RAG semantic search. All 11 Supabase containers run Docker-isolated behind an nginx reverse proxy with Let's Encrypt TLS; Postgres/Auth/Realtime are bound to localhost (only the Kong API gateway is public).
- **Real-time engine:** Supabase Realtime (WebSockets) for live vote broadcasting and presence ("N live", "X/Y answered"), with an HTTP polling fallback for resilience. Load-tested to **300 concurrent participants at 100% success (~1.6 s latency)**; 500 concurrent under extreme synthetic synchronized load reaches 85–94% (real events with natural jitter perform better).
- **AI layer:** Google Gemini via Vercel edge functions. As of July 2026 the platform runs the **latest Gemini 3 family** — **gemini-3.6-flash** (the newest stable flash model) for high-volume generation/grading/insights, **gemini-3.1-pro-preview** for advanced reasoning, and **gemini-embedding-2** (multimodal, 768-dim Matryoshka) for RAG. A daily AI budget guard and per-IP rate limiting protect cost and abuse.
- **SEO as a growth channel:** build-time prerendering + sitemap + structured data make every template and page indexable without client JS — critical for organic discovery in a small-language market.
- **Clean, maintainable codebase:** the two largest views (Presenter, Landing) were refactored from ~1,100 lines each to ~300/240 lines via well-scoped sub-components; the live-critical

Presenter render path is covered by an end-to-end Playwright suite spanning all 8 activity types.

4.3 Data model (PostgreSQL)

Core tables: `events`, `polls`, `options`, `votes`, `questions`, `profiles`, `survey_responses`, `community_templates`, `curriculum_chunks` (pgvector), `organizations`, `org_members`, `org_invites`, `reactions`. Row-Level Security policies enforce per-event and per-organization isolation; anonymous participants vote via session-scoped identifiers.

4.4 Testing & quality

- **End-to-end (Playwright):** live create → vote → Presenter-results flows for all 8 activity types + remote control + landing/join flows.
- **Unit (node:test):** embeddings, SEO helpers, questions core logic.
- **Production verification:** automated build (Vite + sitemap + prerender) on every push via GitHub → Vercel.

5. Traction & Metrics

5.1 Verified production data (July 2026)

Metric	Value
Registered users (auth)	102
Events created	138+
Polls / activities created	249
Curriculum chunks (RAG)	228
Starter templates	39 (13 subjects)
Community templates	19
Supported activity types	8
Concurrent load proven	300 @ 100% success

5.2 Marketing reach (landing page, to be validated with analytics)

- 800+ teachers reached, 12,000+ participant responses, 98% reported satisfaction.
- “Trusted by” strip: UKIM, DUI, FINKI, Gymnasium “Skopje”, MON trainings, corporate trainings.

Honest framing: *MKD Slidea is an early-stage, solo-founder product with strong technical foundations and early organic traction, not yet a scaled business. The investment thesis is the*

regional expansion enabled by completing i18n and enterprise features on top of an already-production-grade platform.

6. Business Model

6.1 Pricing (live)

Plan	Price	Includes
Free	€0	Up to 200 participants/event, all 8 activity types, templates
Pro Monthly	€5 / month	Unlimited participants, branding, exports, AI insights
Pro Yearly	€20 / year	Same as Pro, best value
Institutional	Custom (€200–€2,000+/yr)	Multi-teacher orgs, admin control, centralized billing, white-label option

Billing is currently manual (PayPal, IBAN, MKD bank transfer) with Resend email notifications — a deliberate early-stage choice; automated Stripe/Paddle billing is on the roadmap.

6.2 Revenue streams

1. Pro subscriptions (teachers, trainers).
2. Institutional / school / university licenses (highest value).
3. Future: template marketplace revenue share, API access for LMS integrations, white-label enterprise.

7. Financial Plan (3-Year Projection)

The figures below are illustrative projections based on stated assumptions, not guarantees. They model a conservative regional expansion. All amounts in EUR.

7.1 Current cost structure (lean, monthly)

Item	Cost
Hostinger VPS (self-hosted Supabase)	~€10 (existing)
Vercel Pro	~€20
Gemini API (Tier 1, low volume)	~€0–20
Domain + email (Resend)	~€5
Total	~€35–55 / month

The self-hosted architecture keeps infrastructure cost near-flat as the user base grows initially — a structural margin advantage versus competitors paying per-seat SaaS infrastructure.

7.2 Revenue projection (illustrative)

	Year 1	Year 2	Year 3
Pro subscribers	100	500	2,000
Pro revenue (€20/yr avg)	€2,000	€10,000	€40,000
Institutional licenses	10	50	200
Institutional revenue (€500 avg)	€5,000	€25,000	€100,000
Total ARR	€7,000	€35,000	€140,000
Infrastructure + ops cost	€1,500	€6,000	€20,000
Marketing & sales	€3,000	€15,000	€45,000
Contribution	€2,500	€14,000	€75,000

Assumptions: freemium → Pro conversion ~3–5%; institutional sales driven by direct outreach to schools/universities/NGOs; Year 2–3 growth unlocked by Albanian & Serbian localization (regional TAM expansion ~10×) and enterprise features (SSO, white-label).

7.3 Use of funds (on a €50,000 raise)

Allocation	Amount	Purpose
Regional i18n & localization	€15,000	Complete Albanian, Serbian, Croatian, Bulgarian, Romanian
Cloud infrastructure (scale-out)	€12,000	Highly-available Postgres, Redis cache, CDN, object storage
Enterprise features	€10,000	SSO (SAML), white-label/custom domain, automated billing, API
Marketing & sales	€10,000	Content, comparison SEO, school outreach, demos
Contingency	€3,000	—

8. The Ask — Alibaba Cloud (Combined Proposal)

We propose a **flexible, tiered partnership** — Alibaba Cloud may choose any combination:

Option A — Cloud Infrastructure Support (donation / credits)

As MKD Slidea scales from a single VPS to a regional, highly-available deployment, infrastructure becomes the primary cost and risk. We seek **Alibaba Cloud credits or sponsored infrastructure**: - **ApsaraDB for PostgreSQL** (managed, with vector search) or ECS-hosted Supabase for high availability and automated backups. - **Tair/Redis** for caching and rate limiting; **OSS** for media/template storage; **CDN** for low-latency delivery across the Balkans. - **Model Studio / Qwen API credits** to diversify the AI layer (see Option C).

This removes the single-point-of-failure of the current self-hosted VPS and positions the platform for enterprise-grade SLAs.

Option B — Direct Seed Investment

A **€25,000–€100,000** seed round to execute the 3-year plan (Section 7), in exchange for equity (terms negotiable). Funds allocated per Section 7.3. The investment thesis: a production-grade, low-burn platform with a clear, underserved regional market and a 10× TAM expansion unlocked by localization.

Option C — Strategic Partnership

- **AI integration:** adopt **Alibaba Cloud Qwen / Model Studio** as a complementary (or alternative) LLM provider alongside Google Gemini — reducing vendor concentration, lowering inference cost, and aligning with the Alibaba ecosystem. The architecture is already provider-agnostic at the edge-function layer.

- **Marketplace distribution:** list MKD Slidea on the **Alibaba Cloud Marketplace** to reach Alibaba Cloud’s education and enterprise customers in CEE/Balkans.
- **Joint go-to-market:** co-marketing to Alibaba Cloud’s regional partners; MKD Slidea as a localized edtech reference customer for Alibaba Cloud in Southeast Europe.

Strategic fit for Alibaba Cloud

- A **localized, production-grade edtech entry point** into the Western Balkans — a market global cloud providers serve but rarely localize for.
- A real-world **Qwen / Model Studio reference deployment** in education.
- Alignment with Alibaba Cloud’s EMEA expansion and its open-source / developer-ecosystem strategy.

9. Roadmap (next 12 months)

Critical (pre-scale): - GDPR cookie-consent banner. - Complete i18n: Albanian (sq) → Serbian (sr) → Croatian/Bulgarian/Romanian. - Join-flow validation & error UX.

Q3 2026: - Custom branding UI (color picker + logo upload). - Co-host management UI. - Rate limiting on email/auth endpoints. - Server-side branded PDF export; CSV export button.

Q4 2026 / 2027: - Public REST API + OpenAPI docs (for Moodle, Google Classroom, Microsoft Teams integrations). - Comparison SEO landing pages (vs Mentimeter / Kahoot! / Slido). - Template marketplace advanced filtering. - Enterprise SSO (SAML/OpenID) and white-label / custom domain. - Per-participant analytics; session recording/replay.

10. Risks & Mitigation

Risk	Mitigation
Solo-founder key-person risk	Documented codebase, automated tests, AI-assisted development; investment enables first hires
Self-hosted single point of failure	Option A cloud migration → managed HA Postgres + automated backups
LLM vendor concentration (Google)	Provider-agnostic AI layer; Option C adds Alibaba Qwen
Slow institutional sales cycle	Freemium bottom-up adoption + direct institutional outreach; local invoicing
Small domestic market	Regional (Balkan) expansion via i18n — 10× TAM
Preview-model deprecation (Gemini 3.1 Pro)	Env-driven model config; stable fallbacks; easy model swaps (proven this session)

11. Team

Igor Bogdanoski — Founder & full-stack engineer. Sole developer and product owner; built the entire platform (frontend, self-hosted infrastructure, AI integration, SEO, legal) with AI-assisted engineering. Based in Prilep, North Macedonia. Contact: igorbogdanoski@mismath.net · +389 70 246 814.

12. Appendix

12.1 Live URLs

- Production: <https://slidea.mismath.net>
- Public templates: <https://slidea.mismath.net/templates>
- Self-hosted Supabase API: <https://supabase.mismath.net>

12.2 AI model configuration (current)

Purpose	Model	Notes
Generation / grading / insights	gemini-3.6-flash	Latest stable flash
Advanced reasoning	gemini-3.1-pro-preview	Latest pro (preview)
RAG embeddings	gemini-embedding-2	Multimodal, 768-dim Matryoshka

12.3 Technology index

React 18 · Vite · TailwindCSS · Framer Motion · Zustand · React Router 7 · Recharts · d3-cloud · Supabase (PostgreSQL 16 + pgvector, Auth, Realtime, Storage) · Docker · nginx · Let’s Encrypt · Vercel (edge functions + CDN) · Google Gemini API · satori (OG images) · VAPID web push · Resend email · Playwright (e2e) · node:test (unit).

This document combines technical documentation and a financial/investment proposal. Financial projections (Section 7) are illustrative and assumption-based. Prepared July 2026.